

Pathway ID	Classified pathway chain	Report use
P1	Technically repairable hardware, hinges, handles or fittings → longer service life → physical access blocked by finishes, reveals or installation detailing → full-window replacement despite technical repairability → implementation gap → design for component access, maintenance clearance and documented repair sequence	Use in report section on repairability versus practical implementation
P2	Replaceable components and spare parts → product-life extension → compatible parts unavailable after long use period → complete window replaced earlier than technically necessary → implementation gap → long-term spare-part guarantee and compatibility requirements	Use in report section on supplier commitment and service infrastructure
P3	Replaceable visible or functional components → easier maintenance and targeted upgrading → repair becomes an opportunity for cosmetic or performance upgrading before failure → more frequent replacement of parts or whole windows → potential behavioural rebound → repair-first hierarchy and replacement criteria	Use carefully as a possible rebound pathway, not a proven measured rebound effect
P4	Highly modular or high-performance window system → easier disassembly and better performance → added connectors, materials or technical complexity → higher initial production burden may reduce expected benefit → lifecycle trade-off → minimise added materials and verify through lifecycle comparison	Use in report section on why circular design needs lifecycle verification
P5	Long-life frame → reduced full-window replacement → future glazing or performance upgrades restricted → extended operational inefficiency or disruptive later replacement → technological lock-in → design for future glazing replacement and upgrade compatibility	Use in report section on long-life components and future adaptability
P6	Component repair and maintenance strategy → lower material demand than full replacement → repeated service visits, specialist labour or long-distance transport → benefits partly reduced by transport and intervention burdens → burden shifting → group repairs and support local repair infrastructure	Use in report section on transport and repeated intervention uncertainty
P7	Reuse of removed windows or major components → avoided new production → custom dimensions, certification uncertainty and transport barriers → reuse not realised or net benefit reduced → implementation gap or burden shifting → dimensional standardisation, local reuse networks and documentation	Use in report section explaining why direct reuse was excluded from the main scenario comparison